

SDLC Assignment - 1

SDLC Overview:

(i) Requirement phase

- understand what the client or users need.

activities:

- stakeholder interviews
- surveys or questionnaires
- use case creation.

Importance:

- forms the foundation of the project
- prevents scope creep and miscommunication

(ii) Design phase:

translate requirements into a technical blueprint.

Activities:

- High-level architectural design
- low level detailed design
- UI/UX mockups.

importance:

- sets how the system will work
- identifies tools, languages and architecture needed.

(iii) Implementation (coding) phase:

convert design into working code

Activities:

- Actual development by programmers
- version control and continuous integration.

Importance:

- produce the product
- time and cost intensive, needs good coding practice.

(iv) Testing phase:

ensure the product works as intended.

Activities:

- unit testing
- Integration testing
- User Acceptance Testing (UAT)

Importance:

- Detects bugs before deployment.
- Improves product quality and user Trust.

(v) Deployment phase:

- Release the software to users.

Activities:

- Hosting on servers/cloud
- Monitoring
- Feedback collection

Importance:

- Delivers value to the customer
- Ensures system stability in real environment

Interconnections:

- Feedback from later phase (eg. Testing) can lead to revisiting Design or Requirements.
- Agile SDLC models may loop through these phases multiple times (iterations)